SRI: LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 0$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*}} \left[\sum_{d=0}^{0} F_{s,d} + O(\mathrm{SR}^1)\right].$$

Relative order 0.

$$F_{s,0} = 1$$

1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2} \left[\sum_{d=0}^{0} F_{t,d} + O(\mathrm{SR}^1)\right].$$

Relative order 0.

$$F_{t,0} = 1$$

2 Consistency relations

2.1 $n = 0$

For $p = 0$, the absolute cutoff is $W = 1$:

$$\alpha_0^t = \sum_{d=1}^{1} R_{0,d} + O(\mathrm{SR}^2).$$

Absolute order 1.

$$R_{0,1} = -\frac{r}{8}$$

2.2 $n = 1$

For $p = 0$, the absolute cutoff is $W = 2$:

$$\alpha_1^t = \sum_{d=2}^{2} R_{1,d} + O(\mathrm{SR}^3).$$

Absolute order 2.

$$R_{1,2} = \frac{r^2}{64} + \frac{\alpha_0^s r}{8}$$

2.3 $n = 2$

For $p = 0$, the absolute cutoff is $W = 3$:

$$\alpha_2^t = \sum_{d=3}^{3} R_{2,d} + O(\mathrm{SR}^4).$$

Absolute order 3.

$$R_{2,3} = -\frac{r^3}{256} - \frac{3\alpha_0^s r^2}{64} - \frac{\left(\alpha_0^s\right)^2 r}{8} + \frac{\alpha_1^s r}{8}$$

2.4 $n = 3$

For $p = 0$, the absolute cutoff is $W = 4$:

$$\alpha_3^t = \sum_{d=4}^{4} R_{3,d} + O(\mathrm{SR}^5).$$

Absolute order 4.

$$R_{3,4} = \frac{3r^4}{2048} - \frac{\alpha_1^s r^2}{16} + \frac{\alpha_2^s r}{8} + \frac{\left(\alpha_0^s\right)^3 r}{8} + \frac{3\alpha_0^s r^3}{128} + \frac{7\left(\alpha_0^s\right)^2 r^2}{64} - \frac{3\alpha_0^s \alpha_1^s r}{8}$$